Parth Shingane

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EDUCATION

MIT Academy of Engineering, Pune **2024 – 2028**
B.Tech in Computer Science and Engineering *CGPA: 7.71 / 10.00 | Latest SGPA: 8.09 / 10.00*

SUMMARY

Engineering student who turns real-world problems into working software, with a focus on accessibility and worker safety. Combines full-stack development with AI, IoT and hardware to build solutions that are practical, usable and live. Looking for an opportunity to contribute to a team building products with real impact.

TECHNICAL SKILLS

- **Programming Languages:** Java, C++, Python, JavaScript
- **Backend & Web Development:** Node.js, React.js, Next.js, FastAPI
- **Databases & Tools:** MongoDB, Git, GitHub
- **ML/AI & DataScience:** Machine Learning (Regression, Classification, Clustering), RAG, LightGBM, Random Forest, Seasonal-Naive, A* Algorithm, Model Training & Evaluation, Pandas, NumPy, OKF

KEY PROJECTS

SafeDrain – Sanitation Worker Safety & Monitoring System

– Developed a safety monitoring platform for sanitation workers, integrating real-time hazard monitoring, emergency alerts, and centralized incident tracking.
– Implemented machine learning-based anomaly detection using Isolation Forest and Random Forest to identify potential workplace hazards and support proactive risk assessment.
– **Tech Stack:** React.js, Python, FastAPI, MongoDB, Machine Learning, IoT.

Krishi Bhav: AI-Powered Agricultural Price Intelligence & Market Linkage Platform

– Developed an AI-powered agricultural platform to help farmers make informed selling decisions through market price forecasting and mandi recommendations.
– Integrated agricultural price prediction using historical Agmarknet data with a market linkage workflow to connect farmers with buyers and improve price transparency.
– **Tech Stack:** React.js, Python, FastAPI, MongoDB, Machine Learning (LightGBM), Agmarknet Dataset.

A* Pathfinding Robot: Intelligent Autonomous Navigation System

– Developed an autonomous pathfinding robot using the A* search algorithm to calculate optimal routes from a starting point to a target while avoiding obstacles.
– Implemented heuristic-based path planning using movement costs and estimated distances to enable efficient navigation through grid-based environments.
– **Tech Stack:** Python/C++, A* Search Algorithm, Heuristics, Robotics, Obstacle Detection, Embedded Systems, Computer Vision, Arduino.

ADDITIONAL INFORMATION

- **Achievement:** Participated in national-level hackathons, advancing to the semi-finals and developing innovative, technology-driven solutions to real-world problems.
- **Language:** English, Marathi, Hindi
- **Extra curriculum:**
  1. State-Level Handball & Kabaddi Player
  2. Tech Member of Girl Script Club (MITAOE)
  3. National-Level Hackathon Finalist (SAMVED)
  4. Active participant in technical projects and team-based competitions